Update Joint Modeling 03/11

Overview of the changes

Following changes are made compared to last report:

1. MI-LSPIM results added (up to 100 clusters; MI-LSPIM was not run at 1.000 clusters).
2. MI-LSPIM added to the overview of methods.
3. MSE of β_3 split into bias^2 and variance.

Overview methods

5 different methods are considered & implemented:

- Classical maximum likelihood
- Multiple imputation first (as in Van der Elst #471) followed by applying the closed form (Florez Poveda #546)
- Closed form with iterative reweighting (Florez Poveda #568)
- LSPIM
- Multiple imputation followed by LSPIM (MI-LSPIM)

Table 1: Configuration of the methods; N is the number of clusters (=patients) in the fit.

Method	Configuration
Classical ML	`lme4::lmer`, ML (REML = FALSE); Wald t test on β_3 with $N-2$ df
MI + closed form	`mice 2l.pmm`, 3 imputations, `maxit` = 10; closed form (CbC) per imputation, pooled with Rubin's rules; Wald t test on β_3 with Barnard-Rubin df (complete-data df $N-2$)
Reweighting	Closed form, reweighted until max $\Delta\beta < 10^{-6}$, at most 30 iterations; Wald t test on β_3 with $N-2$ df
LSPIM	Pseudo-score GEE model (Firth-penalised, Ford-Westgate small-sample sandwich); Holm-adjusted test that the per-visit treatment effects differ;
MI-LSPIM	MI as in MI + closed form, followed by LSPIM per imputation; pooled with Rubin's rules.

Table 2: Definition of "not converged" per method.

Method	"Not converged" when
Classical ML	optimizer code $\neq 0$ or an lme4 convergence message (e.g. max|grad| > 0.002)
MI + closed form	never; a failed fit in any imputation counts as an error
Reweighting	the reweighting loop reaches 30 iterations without meeting the $\Delta\beta$ criterion
LSPIM	the penalised GEE does not converge (fitted probabilities numerically 0 or 1)
MI-LSPIM	as LSPIM, in at least one imputation; a failed fit in any imputation counts as an error (as for MI + closed form)

Each replicate gets exactly one of five convergence statuses, checked in this order:

Table 3: Convergence statuses.

Status	Meaning
Error	The fit failed (no estimates); excluded from all metrics.
Not converged	Fit returned estimates, but the method's convergence criterion (table above) was not met.
Converged (singular)	Converged, but the random-effects covariance is at the boundary (smallest eigenvalue $\leq 10^{-6}$, or the final D repaired to positive definite).
Converged (warning)	Converged and not singular, but a warning was raised (e.g. lme4's boundary message at its looser tolerance; for reweighting NaNs in an intermediate iteration; for LSPIM the combined covariance V repaired to the nearest positive semi-definite matrix).
Converged	Converged, not singular, no warning.

- All tests at $\alpha = 0.05$; coverage of the 95% t interval for β_3.
- All statuses except "error" are kept in all metrics.

- LSPIM and MI-LSPIM give only a test decision, no estimate of β_3: they appear in the type I error, power, convergence and computation time results, not in MSE or coverage. Their errors and warnings are described in the appendix.
- MI-LSPIM was not run at $n = 1000$: those cells are shown as "-" in the tables and have no point in the figures.

Simulation setup

Linear model, $t_j = j = 0, \ldots, 11$:

$$y_{ij} = \beta_0 + \beta_1 Treatment_i + \beta_2 t_j + \beta_3 Treatment_i \times t_j + b_{0i} + b_{1i} t_j + \varepsilon_{ij}$$

Log model, $t_j = j = 0, \ldots, 11$:

$$y_{ij} = \beta_0 + \beta_1 Treatment_i + \beta_2 \log(1 + t_j) + \beta_3 Treatment_i \times \log(1 + t_j) + b_{0i} + b_{1i} \log(1 + t_j) + \varepsilon_{ij}$$

In both models $(b_{0i}, b_{1i})^\top \sim N(0, D)$ with $D = \begin{pmatrix} d_{11} & d_{12} \\ d_{12} & d_{22} \end{pmatrix}$ and $\varepsilon_{ij} \sim N(0, \sigma^2)$.

Table 4: Data-generating parameters; a / b / c gives the values of the large, small and null effect.

Effect	Symbol	Linear	Log
Intercept	β_0	2.4562	2.4562
Treatment effect	β_1	0	-2.42 / -0.385 / 0
Time effect	β_2	0.2792	1.2359
Treatment × time	β_3	0.22 / 0.035 / 0	1.948 / 0.310 / 0
Random intercept variance	d_{11}	7.3174	7.3174
Random slope variance	d_{22}	0.2239	4.3875
Random effects covariance	d_{12}	-0.4985	-2.2067
Residual variance	σ^2	3.1508	3.1508

- Linear parameters taken from the toenail data ($\beta_3 = 0.035$); $\beta_3 = 0.22$ added as a larger effect.
- Log parameters: β_2, d_{22} and d_{12} rescaled by $11/\log 12$, so the rise and the slope variance at $t = 11$ match the linear scenario.
- Log crossing: arms cross at $t \approx 2.5$; the treatment difference at $t = 11$ equals that of the linear scenario with $\beta_3 = 0.22$ (large) or 0.035 (small).
- Log null: identical arms ($\beta_1 = \beta_3 = 0$).

Table 5: Simulation design.

Variable	Value
Number of clusters	10, 20, 50, 100, 1000
Number of repeated measures	12
Missingness pattern	"Half-missing" (HM): half of the clusters have no missingness, the other half have a uniform probability to keep 1–11 observations; dropout is monotone. "Three observations minimum" (3O): as half-missing, but with at least three observations per cluster.
Simulations per scenario	5000

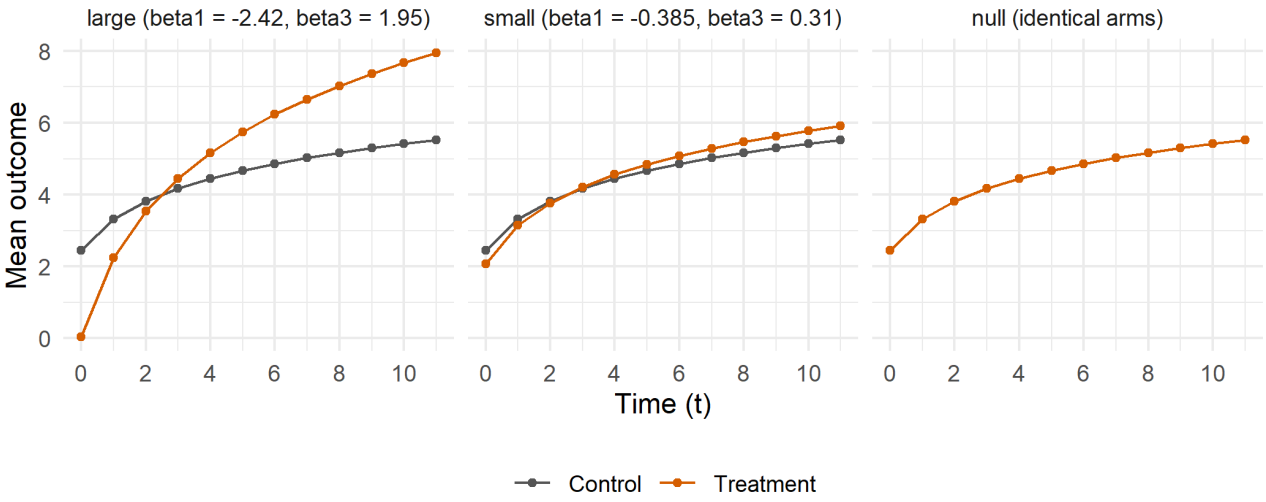


Figure 1: Mean profiles per arm for the log time trend, from the simulation parameters.

Results - linear time trend

MSE, bias² and variance of β_3

MSE = bias² + variance, with bias² = $(\overline{\hat{\beta}}_3 - \beta_3)^2$ and variance = $\frac{1}{R}\sum_r (\hat{\beta}_{3,r} - \overline{\hat{\beta}}_3)^2$ over the R replicates that did not error; the two parts add up to the MSE exactly.

Table 6: MSE, bias² and variance (var) of beta3, half-missing (HM); columns = true beta3: metric.

n	Method	0.22: MSE	0.22: bias²	0.22: var	0.035: MSE	0.035: bias²	0.035: var	0: MSE	0: bias²	0: var
10	Classical ML	1.37e-01	1.28e-05	1.37e-01	1.38e-01	1.36e-04	1.38e-01	1.35e-01	3.37e-05	1.35e-01
10	MI	1.22e-01	7.94e-04	1.21e-01	1.19e-01	1.77e-04	1.19e-01	1.17e-01	1.18e-05	1.17e-01
10	Reweighting	1.36e-01	2.64e-06	1.36e-01	1.38e-01	1.39e-04	1.38e-01	1.34e-01	2.25e-05	1.34e-01
20	Classical ML	6.40e-02	9.02e-06	6.39e-02	6.36e-02	7.47e-08	6.36e-02	6.73e-02	6.47e-05	6.72e-02
20	MI	6.27e-02	6.39e-05	6.26e-02	6.21e-02	3.09e-06	6.21e-02	6.51e-02	4.74e-05	6.51e-02
20	Reweighting	6.50e-02	1.03e-05	6.50e-02	6.43e-02	5.43e-07	6.43e-02	6.84e-02	5.74e-05	6.84e-02
50	Classical ML	2.45e-02	4.73e-06	2.45e-02	2.53e-02	8.14e-06	2.53e-02	2.48e-02	6.62e-07	2.48e-02
50	MI	2.55e-02	1.78e-05	2.55e-02	2.66e-02	1.28e-05	2.66e-02	2.58e-02	1.04e-07	2.58e-02
50	Reweighting	2.49e-02	3.25e-06	2.49e-02	2.56e-02	8.62e-06	2.56e-02	2.52e-02	1.42e-06	2.52e-02
100	Classical ML	1.23e-02	6.79e-06	1.23e-02	1.28e-02	1.59e-08	1.28e-02	1.22e-02	9.10e-07	1.22e-02
100	MI	1.30e-02	6.58e-09	1.30e-02	1.34e-02	3.72e-07	1.34e-02	1.31e-02	8.68e-07	1.31e-02
100	Reweighting	1.25e-02	5.95e-06	1.25e-02	1.29e-02	4.42e-09	1.29e-02	1.24e-02	1.11e-06	1.24e-02
1000	Classical ML	1.29e-03	6.07e-10	1.29e-03	1.25e-03	6.53e-07	1.25e-03	1.25e-03	2.38e-07	1.25e-03
1000	MI	1.41e-03	7.66e-08	1.41e-03	1.35e-03	4.67e-07	1.35e-03	1.36e-03	4.10e-07	1.36e-03
1000	Reweighting	1.32e-03	8.29e-09	1.32e-03	1.27e-03	5.43e-07	1.26e-03	1.27e-03	2.13e-07	1.27e-03

Table 7: MSE, bias² and variance (var) of beta3, three observations minimum (3O); columns = true beta3: metric.

n	Method	0.22: MSE	0.22: bias²	0.22: var	0.035: MSE	0.035: bias²	0.035: var	0: MSE	0: bias²	0: var
10	Classical ML	1.25e-01	7.38e-05	1.25e-01	1.25e-01	7.89e-07	1.25e-01	1.26e-01	2.95e-06	1.26e-01
10	MI	1.13e-01	9.99e-04	1.12e-01	1.10e-01	2.16e-05	1.10e-01	1.11e-01	4.92e-07	1.11e-01
10	Reweighting	1.24e-01	5.99e-05	1.24e-01	1.25e-01	1.35e-06	1.25e-01	1.25e-01	9.92e-07	1.25e-01
20	Classical ML	5.95e-02	3.04e-05	5.94e-02	6.31e-02	1.42e-04	6.29e-02	6.27e-02	3.64e-05	6.26e-02
20	MI	5.88e-02	3.20e-04	5.85e-02	6.06e-02	1.04e-04	6.05e-02	6.08e-02	2.88e-05	6.08e-02
20	Reweighting	5.94e-02	2.52e-05	5.94e-02	6.30e-02	1.50e-04	6.29e-02	6.26e-02	3.72e-05	6.26e-02
50	Classical ML	2.41e-02	1.04e-05	2.41e-02	2.32e-02	2.04e-06	2.32e-02	2.36e-02	4.13e-06	2.36e-02
50	MI	2.54e-02	8.37e-05	2.53e-02	2.40e-02	1.62e-07	2.40e-02	2.45e-02	4.52e-06	2.45e-02
50	Reweighting	2.41e-02	1.05e-05	2.41e-02	2.32e-02	2.14e-06	2.32e-02	2.36e-02	4.10e-06	2.36e-02
100	Classical ML	1.19e-02	1.64e-06	1.19e-02	1.17e-02	1.23e-07	1.17e-02	1.15e-02	2.56e-06	1.14e-02
100	MI	1.26e-02	4.13e-06	1.26e-02	1.23e-02	1.30e-08	1.23e-02	1.21e-02	2.38e-06	1.21e-02
100	Reweighting	1.19e-02	1.62e-06	1.19e-02	1.18e-02	1.00e-07	1.18e-02	1.15e-02	2.57e-06	1.14e-02
1000	Classical ML	1.17e-03	6.61e-07	1.17e-03	1.19e-03	3.49e-09	1.19e-03	1.21e-03	6.61e-08	1.21e-03
1000	MI	1.26e-03	1.63e-06	1.26e-03	1.31e-03	5.71e-10	1.31e-03	1.31e-03	1.23e-07	1.31e-03
1000	Reweighting	1.17e-03	6.64e-07	1.17e-03	1.19e-03	3.60e-09	1.19e-03	1.21e-03	6.43e-08	1.21e-03

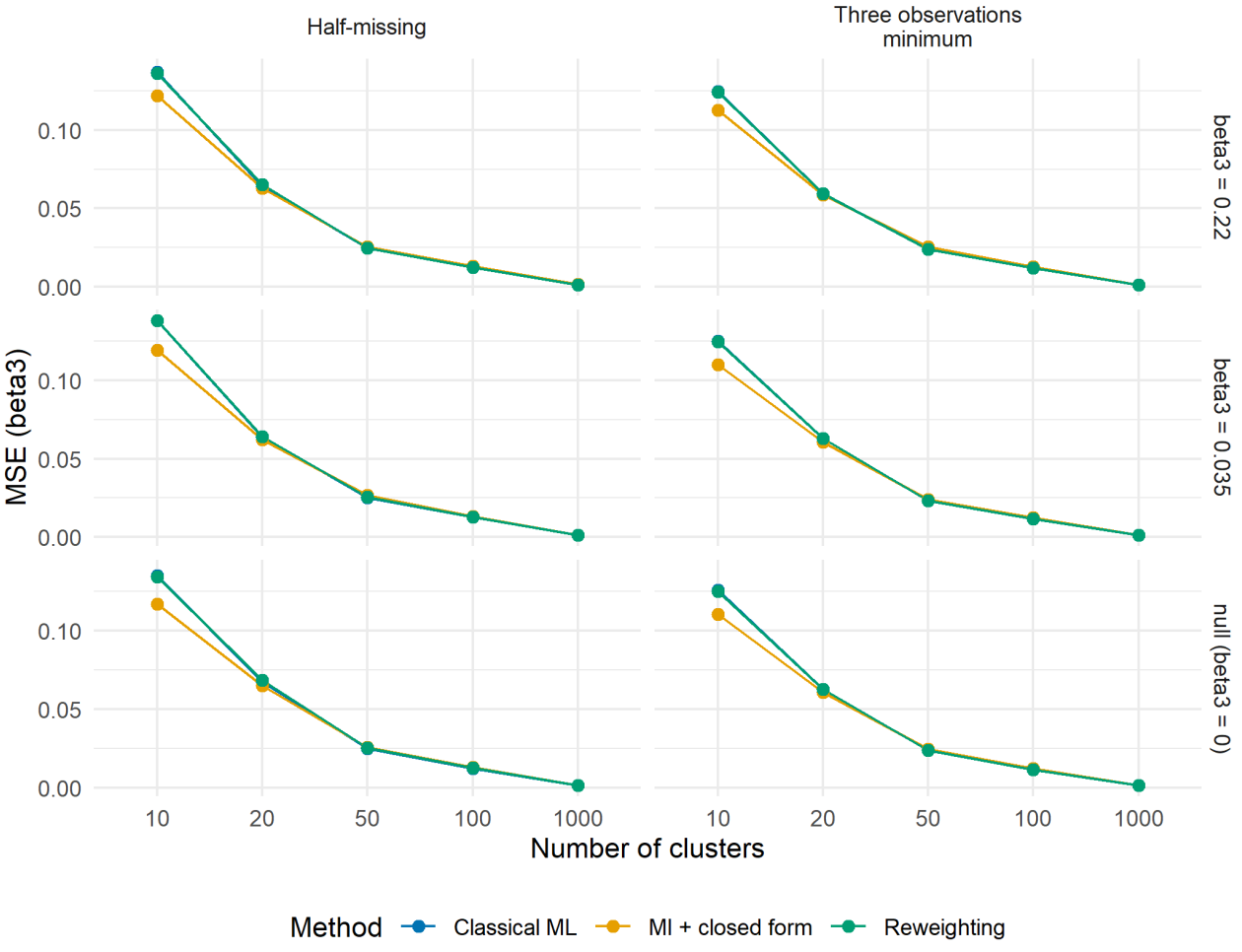


Figure 2: MSE of beta3 for the linear time trend.

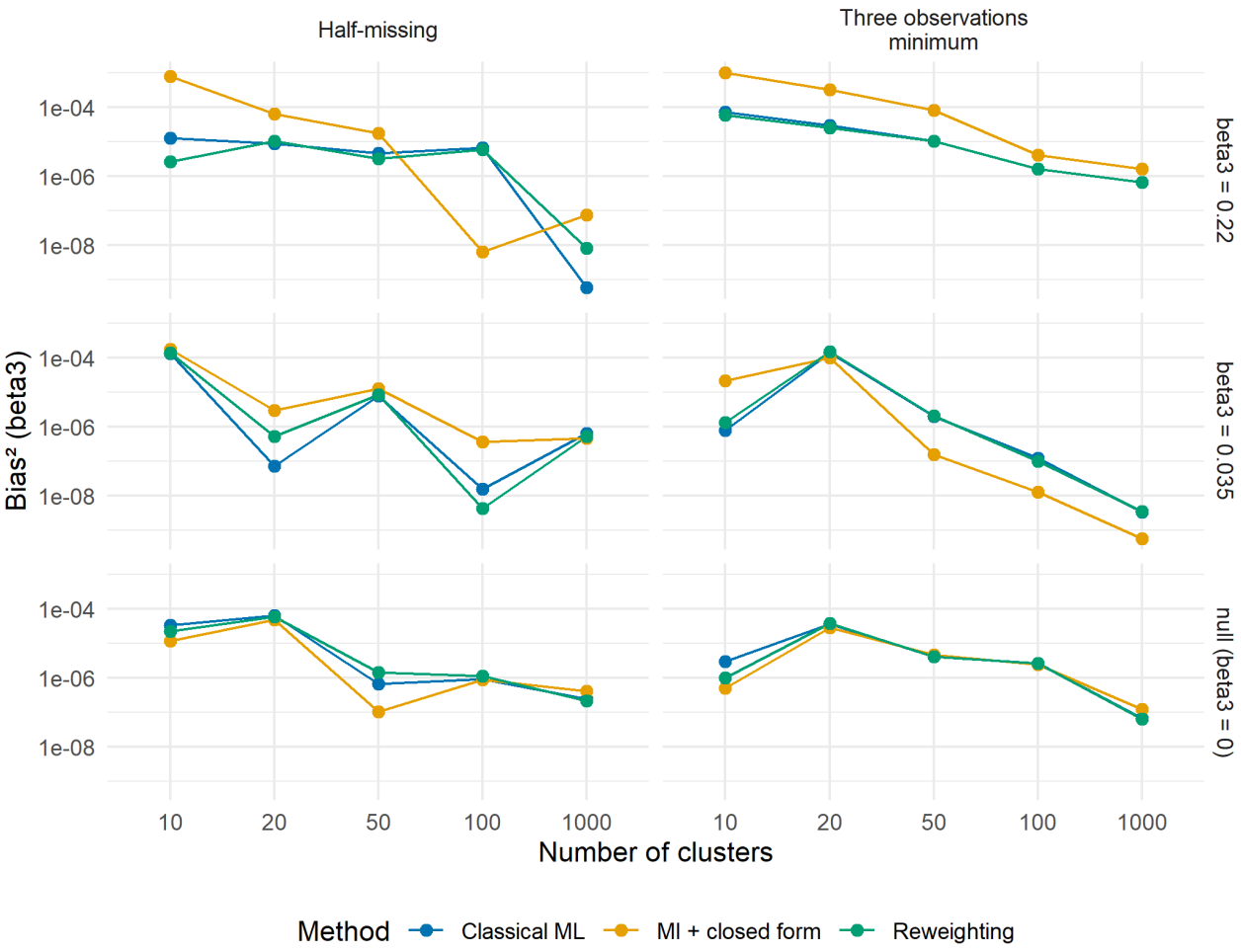


Figure 3: Squared bias of beta3 for the linear time trend (log scale).

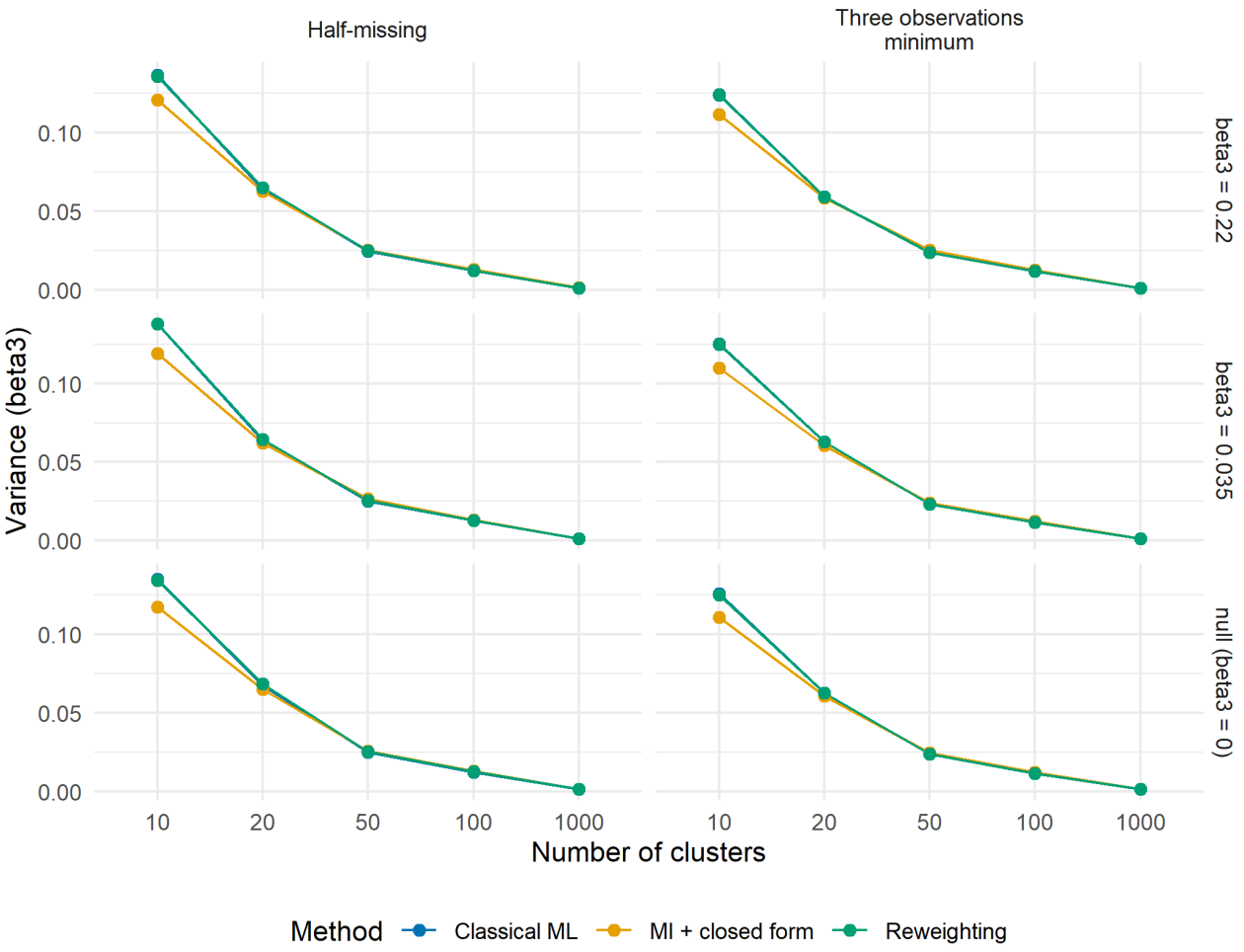


Figure 4: Variance of beta3 for the linear time trend.

- MSE falls with the number of clusters at a similar rate for all methods; the effect size doesn't matter, and MSE is slightly lower under three observations minimum.
- Classical ML and reweighting are nearly identical.
- MI has the lowest MSE at $n = 10$ and 20; from $n = 50$ onwards classical ML and reweighting are slightly better.
- Variance makes up essentially all of the MSE: bias² is at most 0.9% of the MSE in every scenario (bias² between 6e-10 and 1.0e-3), so the differences between methods are differences in variance.
- Most of the bias² is Monte Carlo noise: with $R = 5000$ the standard error of the mean estimate is about 0.005 at $n = 10$, which alone gives a bias² of about 2.5e-5.
- The exception is MI at $n = 10$ and $\beta_3 = 0.22$: bias² is 7.9e-04 under half-missing and 1.0e-03 under three observations minimum (bias about 0.028 and 0.032, 5.7 and 6.7 Monte Carlo standard errors). The bias is still above 4 standard errors at $n = 20$ and 50 under three observations minimum, but its share of the MSE stays below 0.6%.

Type I error

Table 8: Type I error rate of the interaction test (beta3 = 0).

n	Method	HM: 0	3O: 0
10	Classical ML	0.102	0.0988
10	MI + closed form	0.0466	0.0392
10	Reweighting	0.0596	0.0648
10	LSPIM	0.109	0.123
10	MI-LSPIM	0.00260	0.00220
20	Classical ML	0.0780	0.0722
20	MI + closed form	0.0518	0.0490
20	Reweighting	0.0610	0.0606
20	LSPIM	0.0775	0.0813
20	MI-LSPIM	0.00300	0.00520
50	Classical ML	0.0562	0.0546
50	MI + closed form	0.0404	0.0410
50	Reweighting	0.0500	0.0512
50	LSPIM	0.0372	0.0374
50	MI-LSPIM	0.00660	0.00980
100	Classical ML	0.0532	0.0528
100	MI + closed form	0.0430	0.0400
100	Reweighting	0.0500	0.0496
100	LSPIM	0.0370	0.0352
100	MI-LSPIM	0.0102	0.0122
1000	Classical ML	0.0472	0.0494
1000	MI + closed form	0.0472	0.0460
1000	Reweighting	0.0478	0.0488
1000	LSPIM	0.0398	0.0446
1000	MI-LSPIM	-	-

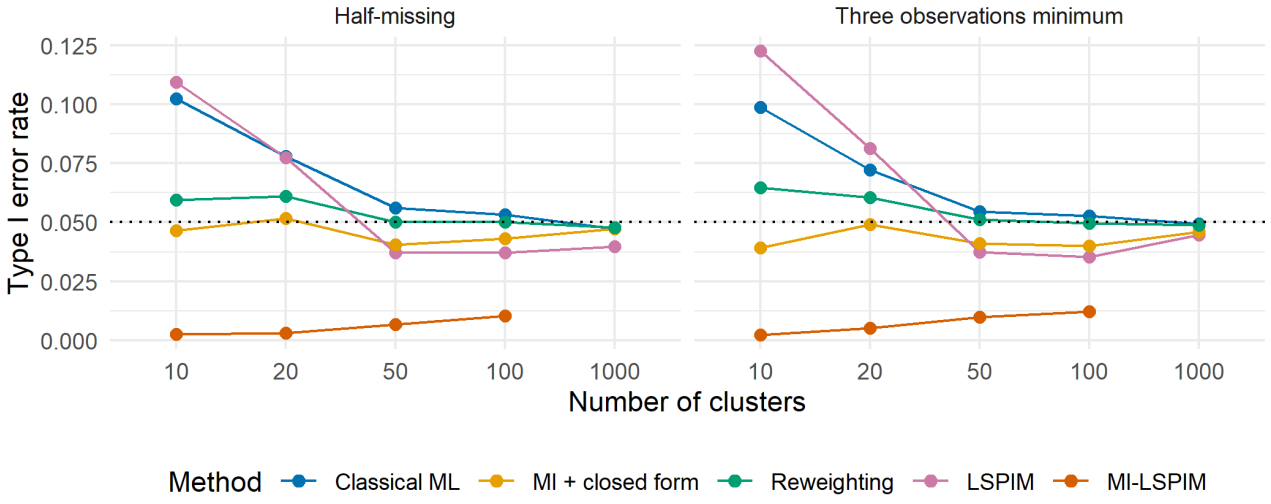


Figure 5: Type I error rate for the linear time trend (dotted line: nominal 0.05).

- Classical ML is anti-conservative at small n (about 0.10 at $n = 10$, 0.07–0.08 at $n = 20$) and close to nominal from $n = 50$.
- Reweighting is slightly inflated at $n = 10$ and 20 (about 0.06) and nominal from $n = 50$.
- MI is at or slightly below nominal at every n (about 0.04–0.05).
- LSPIM is the most inflated at $n = 10$ (0.11–0.12) and still at $n = 20$ (about 0.08); from $n = 50$ onwards it is conservative (0.035–0.045).
- MI-LSPIM is very conservative: 0.002–0.005 at $n = 10$ and 20, rising to 0.010–0.012 at $n = 100$ (LSPIM: 0.035–0.037). It is not run at $n = 1000$.

Power

Table 9: Power of the interaction test.

n	Method	HM: 0.22	HM: 0.035	3O: 0.22	3O: 0.035
10	Classical ML	0.155	0.105	0.140	0.100
10	MI + closed form	0.0708	0.0436	0.0596	0.0372
10	Reweighting	0.0930	0.0626	0.0963	0.0635
10	LSPIM	0.120	0.117	0.123	0.126
10	MI-LSPIM	0.00360	0.00160	0.00400	0.00200
20	Classical ML	0.165	0.0702	0.165	0.0756
20	MI + closed form	0.112	0.0472	0.108	0.0446
20	Reweighting	0.141	0.0574	0.140	0.0650
20	LSPIM	0.130	0.0768	0.119	0.0773
20	MI-LSPIM	0.0114	0.00440	0.0100	0.00520
50	Classical ML	0.290	0.0648	0.295	0.0568
50	MI + closed form	0.220	0.0546	0.215	0.0422
50	Reweighting	0.267	0.0586	0.282	0.0528
50	LSPIM	0.103	0.0394	0.104	0.0414
50	MI-LSPIM	0.0492	0.00820	0.0568	0.0132
100	Classical ML	0.522	0.0654	0.528	0.0642
100	MI + closed form	0.397	0.0582	0.416	0.0486
100	Reweighting	0.500	0.0616	0.518	0.0616
100	LSPIM	0.181	0.0434	0.197	0.0462
100	MI-LSPIM	0.128	0.0164	0.148	0.0182
1000	Classical ML	1.00	0.162	1.00	0.174
1000	MI + closed form	0.998	0.130	1.00	0.142
1000	Reweighting	1.00	0.158	1.00	0.173
1000	LSPIM	0.990	0.0726	0.991	0.0742
1000	MI-LSPIM	-	-	-	-

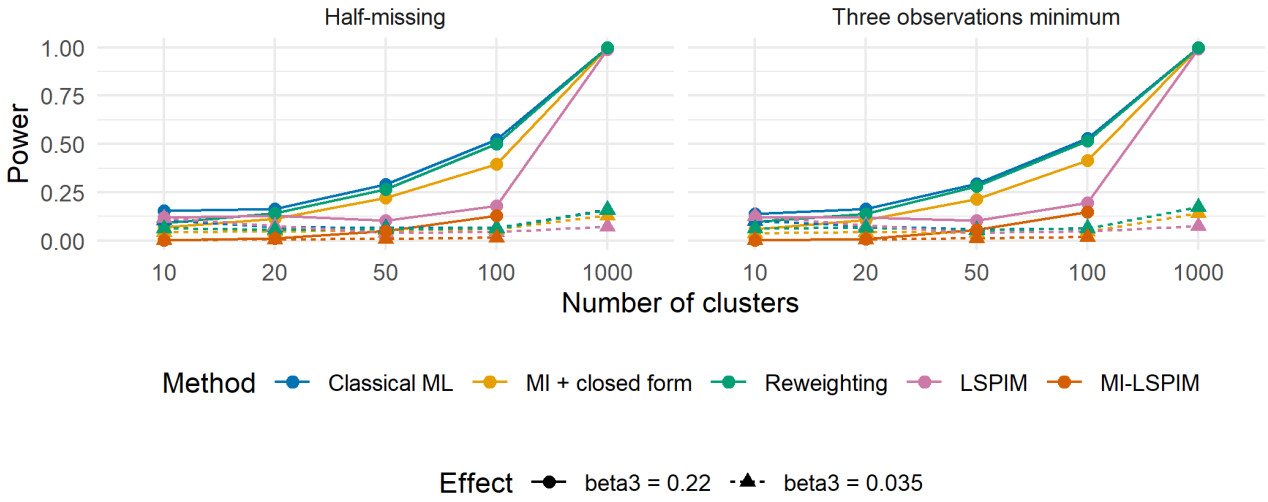


Figure 6: Power for the linear time trend.

- Classical ML has the highest power, then reweighting, then MI; at small n the lead of classical ML partly reflects its type I error inflation.
- For $\beta_3 = 0.22$ classical ML, reweighting and MI reach full power at $n = 1000$; for $\beta_3 = 0.035$ power stays low even at $n = 1000$ (0.13–0.17).
- LSPIM has much lower power than the other methods for a linear effect: for $\beta_3 = 0.22$ about 0.18–0.20 at $n = 100$ (others 0.40–0.53).
- MI-LSPIM has almost no power at small n and is the least powerful method up to $n = 100$: for $\beta_3 = 0.22$ power is 0.004–0.011 at $n = 10$ and 20 and 0.13–0.15 at $n = 100$ (LSPIM: 0.18–0.20); for $\beta_3 = 0.035$ it stays at or below 0.02 at $n = 100$.

Coverage of β_3

Table 10: Coverage of the 95% t interval for beta3.

n	Method	HM: 0.22	HM: 0.035	HM: 0	3O: 0.22	3O: 0.035	3O: 0
10	Classical ML	0.896	0.895	0.898	0.907	0.904	0.901
10	MI + closed form	0.956	0.955	0.953	0.961	0.963	0.961
10	Reweighting	0.940	0.939	0.940	0.942	0.937	0.935
20	Classical ML	0.926	0.930	0.922	0.935	0.924	0.928
20	MI + closed form	0.952	0.954	0.948	0.959	0.959	0.951
20	Reweighting	0.939	0.947	0.939	0.945	0.937	0.939
50	Classical ML	0.948	0.939	0.944	0.943	0.949	0.945
50	MI + closed form	0.956	0.951	0.960	0.953	0.960	0.959
50	Reweighting	0.950	0.946	0.950	0.944	0.952	0.949
100	Classical ML	0.948	0.946	0.947	0.945	0.945	0.947
100	MI + closed form	0.958	0.950	0.957	0.958	0.957	0.960
100	Reweighting	0.952	0.949	0.950	0.946	0.947	0.950
1000	Classical ML	0.950	0.952	0.953	0.956	0.947	0.951
1000	MI + closed form	0.952	0.960	0.953	0.962	0.957	0.954
1000	Reweighting	0.950	0.953	0.952	0.956	0.947	0.951

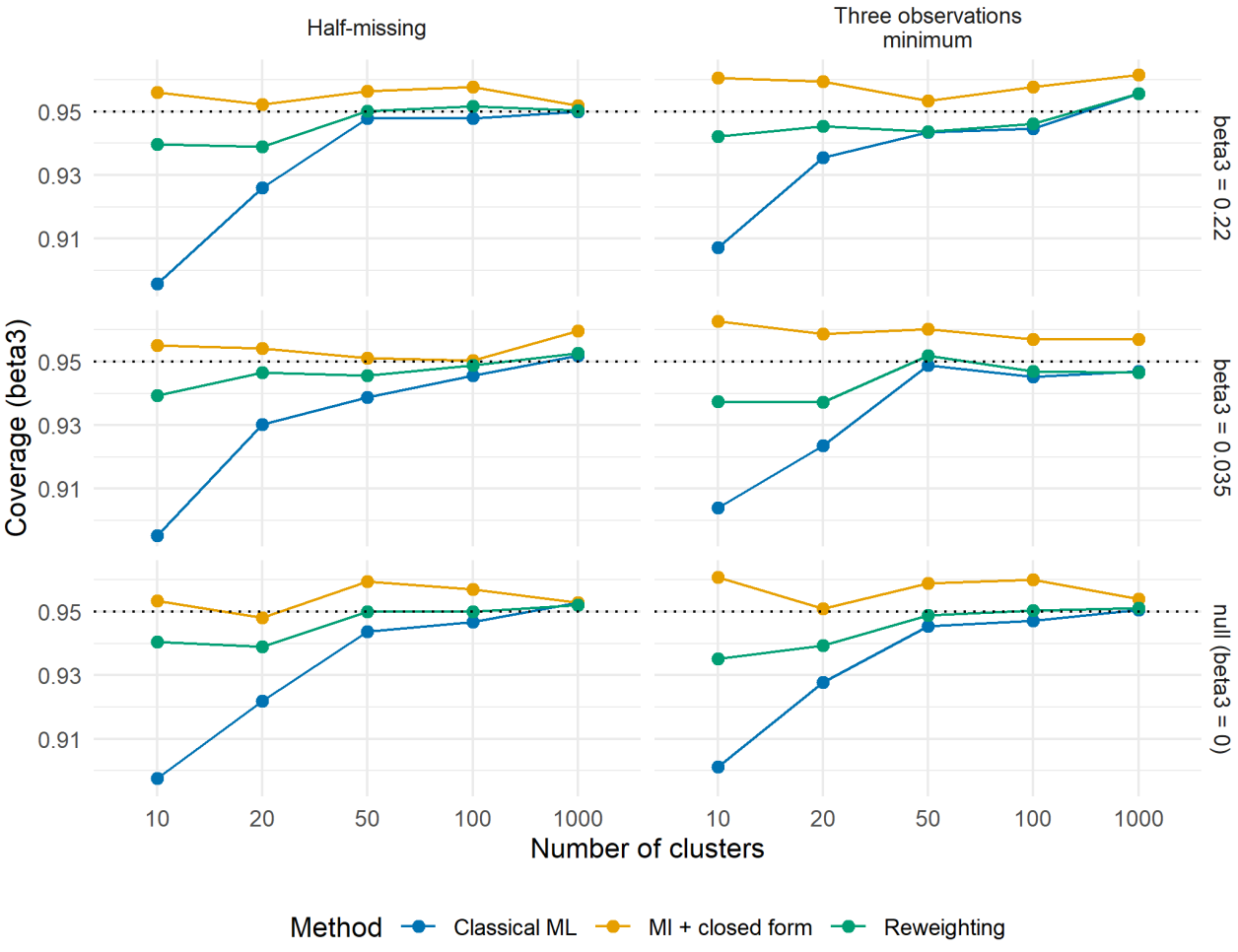


Figure 7: Coverage of the 95% interval for beta3 for the linear time trend (dotted line: 0.95).

- Classical ML has low coverage at small n (about 0.90 at $n = 10$, 0.92–0.93 at $n = 20$) and is close to nominal from $n = 50$.
- Reweighting is slightly below nominal at $n = 10$ and 20 (0.94) and nominal from $n = 50$.
- MI is at or slightly above nominal at every n (0.95–0.96).

Convergence

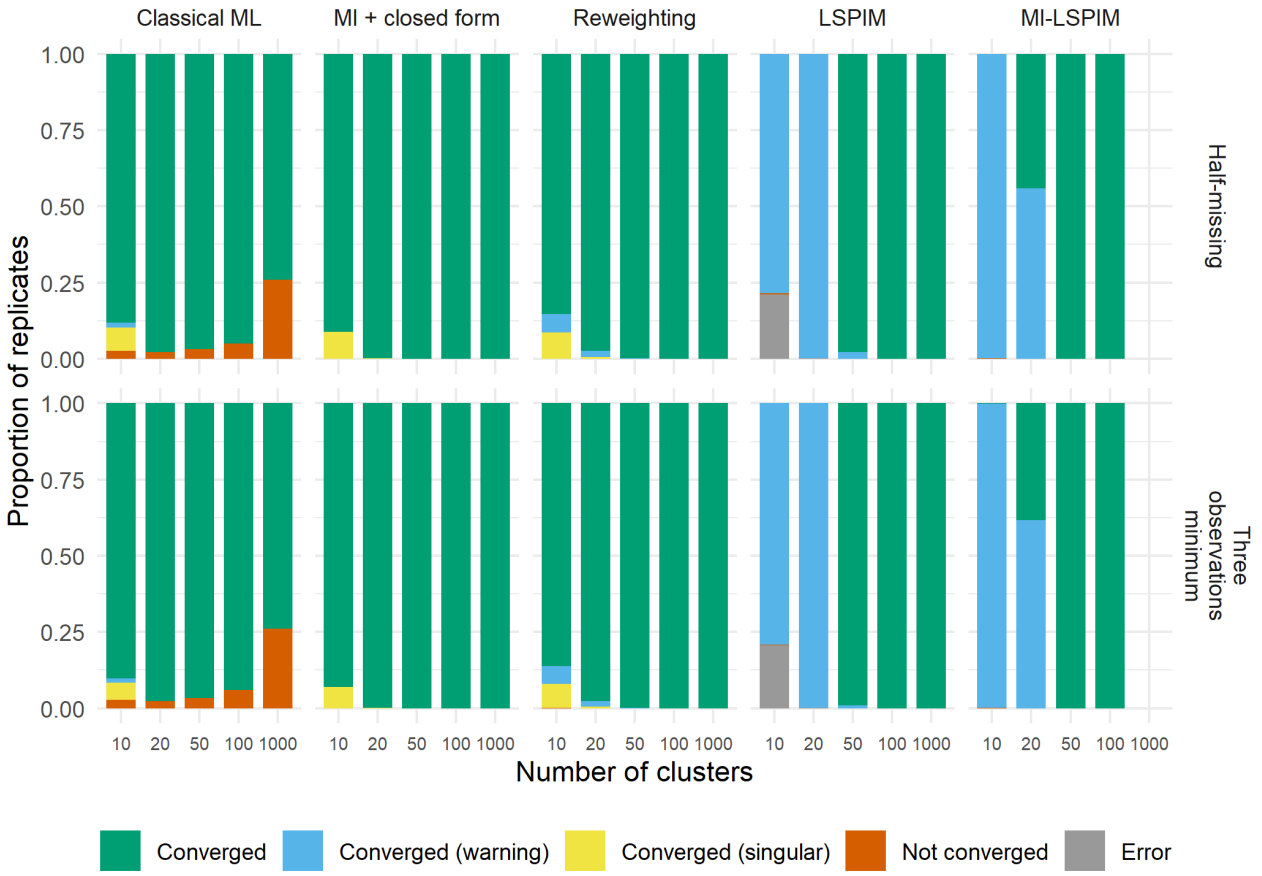


Figure 8: Convergence status for the linear time trend, pooled over effect sizes.

- Singular fits occur at $n = 10$ for classical ML, MI and reweighting at a similar rate (5–9%) and almost vanish from $n = 20$.
- Reweighting and MI essentially always converge; reweighting has a small warning share at $n = 10$ and 20.
- Classical ML has a small "not converged" share at every n, which grows to about 26% at $n = 1000$. This is a false positive of lme4's absolute gradient check, which scales with the sample size: the flagged fits are kept in all metrics.
- LSPIM: about 20% errors at $n = 10$, and almost all fits at $n = 10$ and 20 carry a warning; see the appendix.
- MI-LSPIM has no errors. Nearly all fits at $n = 10$ (99.7%) and more than half at $n = 20$ (about 57%) carry a warning (the combined covariance V repaired to a positive semi-definite matrix); almost none from $n = 50$. "Not converged" is rare (0.2% at $n = 10$, 0.05% over all fits); see the appendix.

Computation time

Table 11: Median computation time per replicate (seconds).

n	Method	HM: 0.22	HM: 0.035	HM: 0	3O: 0.22	3O: 0.035	3O: 0
10	Classical ML	0.0180	0.0170	0.0180	0.0160	0.0160	0.0170
10	MI	1.16	1.04	1.04	1.05	1.05	1.04
10	Reweighting	0.0560	0.0390	0.0430	0.0490	0.0470	0.0500
10	LSPIM	0.0150	0.00800	0.00800	0.00900	0.00900	0.00900
10	MI-LSPIM	1.16	1.17	1.17	1.17	1.18	1.17
20	Classical ML	0.0220	0.0150	0.0180	0.0200	0.0190	0.0200
20	MI	1.26	1.14	1.14	1.15	1.15	1.15
20	Reweighting	0.111	0.0890	0.0900	0.102	0.103	0.102
20	LSPIM	0.0420	0.0130	0.0130	0.0140	0.0140	0.0140
20	MI-LSPIM	1.29	1.29	1.29	1.31	1.30	1.30
50	Classical ML	0.0260	0.0180	0.0210	0.0225	0.0180	0.0190
50	MI	1.58	1.44	1.43	1.45	1.45	1.45
50	Reweighting	0.288	0.231	0.234	0.267	0.264	0.266
50	LSPIM	0.234	0.0490	0.0490	0.0590	0.0570	0.0550
50	MI-LSPIM	1.81	1.79	1.78	1.83	1.80	1.80
100	Classical ML	0.0340	0.0300	0.0290	0.0290	0.0300	0.0290
100	MI	2.17	1.97	1.96	1.99	2.00	2.00
100	Reweighting	0.609	0.511	0.509	0.570	0.569	0.572
100	LSPIM	1.26	0.218	0.214	0.263	0.253	0.251
100	MI-LSPIM	3.63	3.56	3.56	3.65	3.57	3.57
1000	Classical ML	0.285	0.219	0.220	0.230	0.230	0.231
1000	MI	17.1	13.1	13.7	13.5	13.6	13.6
1000	Reweighting	15.8	14.7	14.7	16.4	16.4	16.4
1000	LSPIM	52.1	45.9	43.0	57.6	50.5	47.9
1000	MI-LSPIM	-	-	-	-	-	-

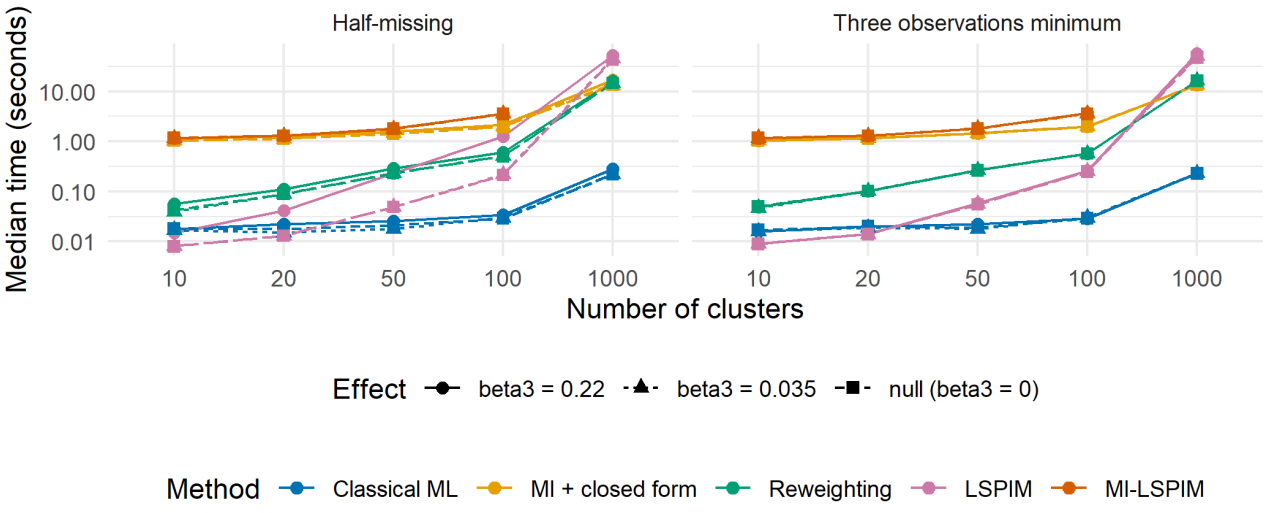


Figure 9: Median computation time per replicate for the linear time trend (log scale).

- Classical ML is fastest from $n = 50$ onwards; at $n = 10$ and 20 LSPIM is slightly faster.
- Of the four earlier methods, MI is the slowest up to $n = 100$ (1–2 seconds per replicate).
- LSPIM is fast up to $n = 100$ (at most about 1 second) but grows roughly quadratically with n: at $n = 1000$ it is the slowest method (about 45–58 seconds), against about 14–17 seconds for MI and reweighting.
- MI-LSPIM takes about 1.2 seconds per replicate at $n = 10$ and about 3.6 seconds at $n = 100$: slower than MI, and the slowest method up to $n = 100$ (LSPIM: at most about 1 second).

Results - log time trend

- MSE and coverage are not reported: the linear methods estimate a linear slope, which has no true value under the log trend.
- Effect sizes: large ($\beta_1 = -2.42$, $\beta_3 = 1.95$), small ($\beta_1 = -0.385$, $\beta_3 = 0.31$).

Type I error

Table 12: Type I error rate of the interaction test (identical arms).

n	Method	HM: 0	3O: 0
10	Classical ML	0.103	0.0976
10	MI + closed form	0.0392	0.0366
10	Reweighting	0.0496	0.0553
10	LSPIM	0.110	0.107
10	MI-LSPIM	0.00140	0.00160
20	Classical ML	0.0762	0.0684
20	MI + closed form	0.0536	0.0454
20	Reweighting	0.0482	0.0430
20	LSPIM	0.0832	0.0731
20	MI-LSPIM	0.00460	0.00380
50	Classical ML	0.0616	0.0560
50	MI + closed form	0.0508	0.0410
50	Reweighting	0.0436	0.0416
50	LSPIM	0.0390	0.0408
50	MI-LSPIM	0.00600	0.0100
100	Classical ML	0.0560	0.0512
100	MI + closed form	0.0478	0.0440
100	Reweighting	0.0386	0.0354
100	LSPIM	0.0406	0.0406
100	MI-LSPIM	0.0120	0.0138
1000	Classical ML	0.0498	0.0456
1000	MI + closed form	0.0432	0.0414
1000	Reweighting	0.0350	0.0338
1000	LSPIM	0.0422	0.0438
1000	MI-LSPIM	-	-



Figure 10: Type I error rate for the log time trend (dotted line: nominal 0.05).

- Classical ML and LSPIM show the same small-n inflation as in the linear scenarios (about 0.10 and 0.11 at $n = 10$); LSPIM is again conservative from $n = 50$.
- MI is at or slightly below nominal at every n.
- Reweighting is close to nominal at $n = 10$ and 20, and becomes conservative from $n = 50$ onwards (about 0.035 at $n = 1000$).
- MI-LSPIM is again very conservative: 0.001–0.005 at $n = 10$ and 20, 0.012–0.014 at $n = 100$.

Power

Table 13: Power of the interaction test.

n	Method	HM: large	HM: small	3O: large	3O: small
10	Classical ML	0.283	0.102	0.285	0.0902
10	MI + closed form	0.134	0.0414	0.141	0.0352
10	Reweighting	0.165	0.0482	0.183	0.0505
10	LSPIM	0.188	0.110	0.218	0.117
10	MI-LSPIM	0.0104	0.00260	0.0176	0.00340
20	Classical ML	0.444	0.0806	0.458	0.0776
20	MI + closed form	0.315	0.0520	0.328	0.0472
20	Reweighting	0.349	0.0518	0.369	0.0556
20	LSPIM	0.254	0.0845	0.238	0.0776
20	MI-LSPIM	0.0714	0.00400	0.0840	0.00580
50	Classical ML	0.783	0.0814	0.812	0.0832
50	MI + closed form	0.643	0.0638	0.678	0.0610
50	Reweighting	0.715	0.0612	0.757	0.0604
50	LSPIM	0.523	0.0490	0.546	0.0482
50	MI-LSPIM	0.470	0.0160	0.516	0.0164
100	Classical ML	0.972	0.103	0.979	0.0946
100	MI + closed form	0.902	0.0798	0.921	0.0722
100	Reweighting	0.959	0.0762	0.969	0.0708
100	LSPIM	0.881	0.0576	0.901	0.0572
100	MI-LSPIM	0.891	0.0224	0.909	0.0276
1000	Classical ML	1.00	0.511	1.00	0.543
1000	MI + closed form	1.00	0.409	1.00	0.426
1000	Reweighting	1.00	0.458	1.00	0.487
1000	LSPIM	1.00	0.288	1.00	0.293
1000	MI-LSPIM	-	-	-	-

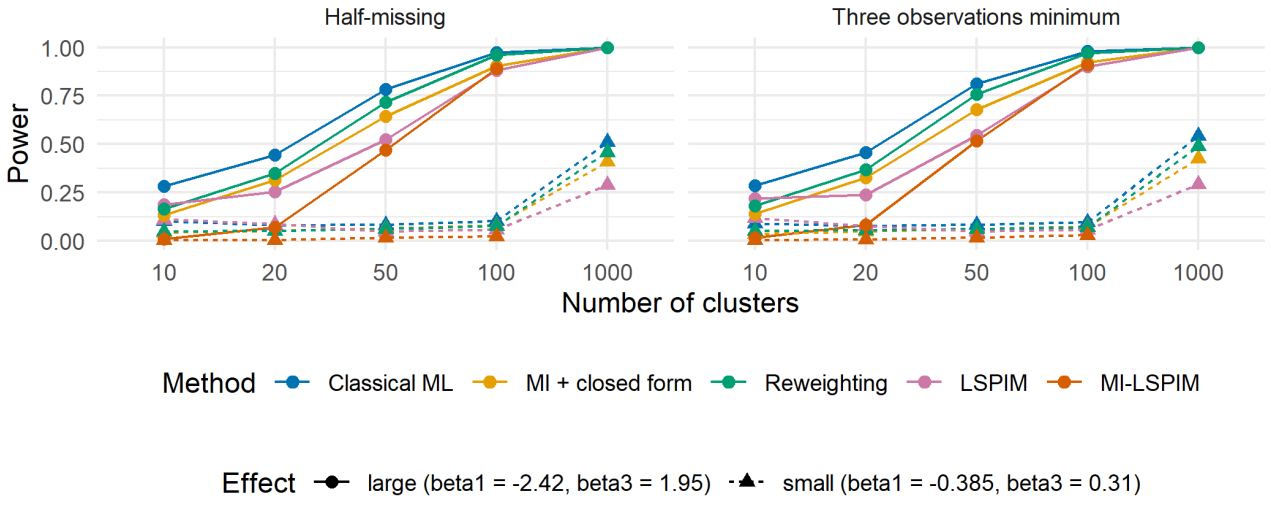


Figure 11: Power for the log time trend.

- The linear methods still detect the interaction effect well: power is higher than in the linear scenarios with the same end-of-study difference (large vs. $\beta_3 = 0.22$, small vs. $\beta_3 = 0.035$).
- Classical ML has the highest power, then reweighting, then MI.
- LSPIM is close to MI for the large effect (about 0.88–0.90 at $n = 100$), but has the lowest power for the small effect (about 0.29 at $n = 1000$, others 0.41–0.54).
- MI-LSPIM is as powerful as LSPIM for the large effect at $n = 100$ (0.89–0.91 against 0.88–0.90) but is behind it at $n = 50$ (0.47–0.52 against 0.52–0.55) and has almost no power at $n = 10$ and 20 (0.01–0.08). For the small effect it has the lowest power (0.02–0.03 at $n = 100$, others 0.06–0.10).

Convergence

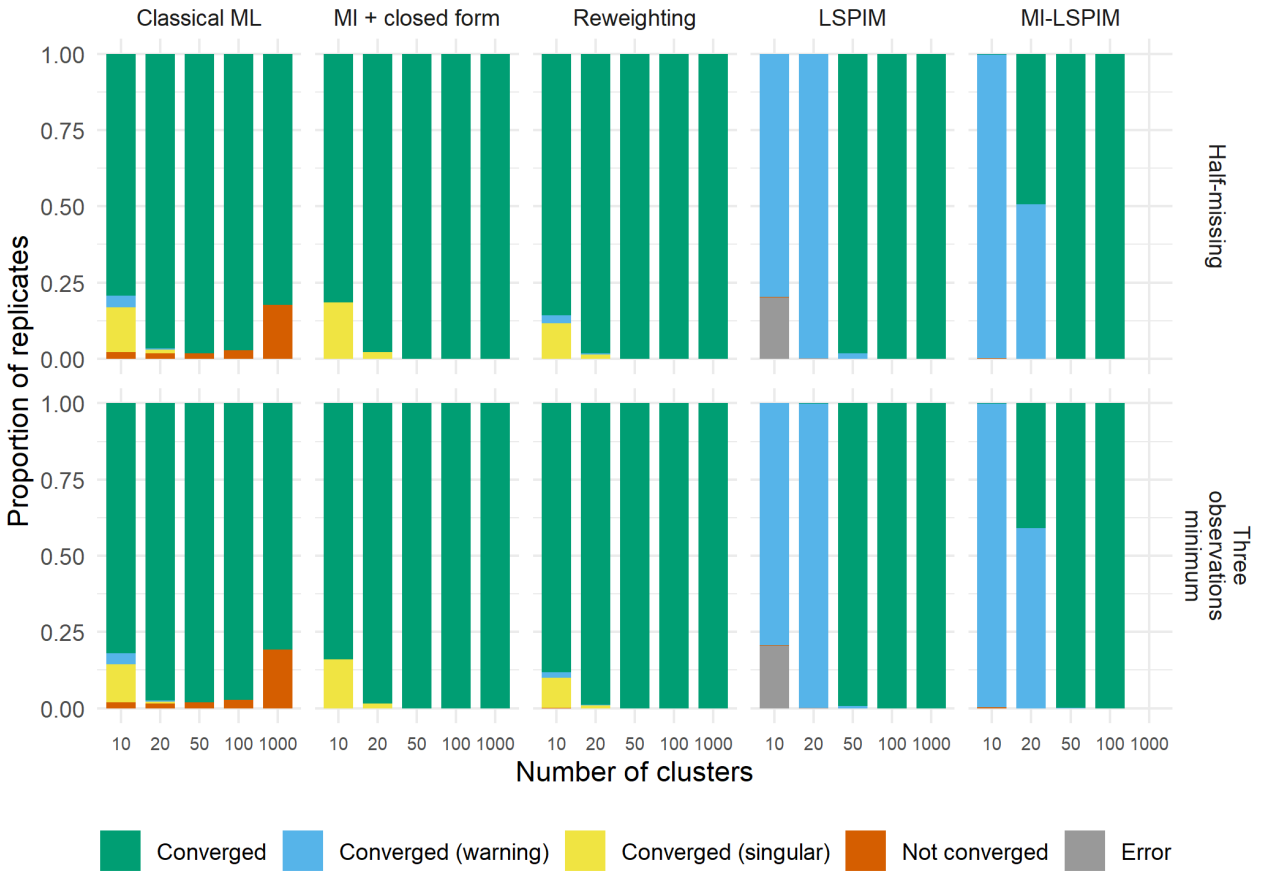


Figure 12: Convergence status for the log time trend, pooled over effect sizes.

- Same pattern as the linear scenarios, with more singular fits at $n = 10$ (10–20%).
- Classical ML's "not converged" share at $n = 1000$ (about 18%) is again the lme4 gradient false positive.
- LSPIM errors and warnings as in the linear scenarios; see the appendix.
- MI-LSPIM: same pattern as in the linear scenarios (no errors; warnings in 99.7% of fits at $n = 10$ and about 55% at $n = 20$).

Computation time

Table 14: Median computation time per replicate (seconds).

n	Method	HM: large	HM: small	HM: 0	3O: large	3O: small	3O: 0
10	Classical ML	0.0170	0.0170	0.0160	0.0140	0.0170	0.0160
10	MI	1.03	1.03	1.02	1.03	1.02	1.03
10	Reweighting	0.0410	0.0390	0.0430	0.0450	0.0470	0.0435
10	LSPIM	0.00800	0.00800	0.00800	0.00900	0.00900	0.00900
10	MI-LSPIM	1.15	1.15	1.15	1.16	1.16	1.15
20	Classical ML	0.0170	0.0200	0.0190	0.0190	0.0190	0.0150
20	MI	1.13	1.13	1.12	1.13	1.13	1.13
20	Reweighting	0.0900	0.0900	0.0900	0.102	0.101	0.102
20	LSPIM	0.0130	0.0130	0.0130	0.0120	0.0140	0.0140
20	MI-LSPIM	1.27	1.27	1.27	1.28	1.27	1.27
50	Classical ML	0.0210	0.0220	0.0170	0.0220	0.0220	0.0210
50	MI	1.41	1.41	1.40	1.41	1.42	1.41
50	Reweighting	0.232	0.230	0.231	0.265	0.263	0.265
50	LSPIM	0.0520	0.0480	0.0480	0.0600	0.0560	0.0560
50	MI-LSPIM	1.79	1.74	1.74	1.80	1.76	1.75
100	Classical ML	0.0280	0.0270	0.0270	0.0290	0.0280	0.0280
100	MI	1.91	1.90	1.89	1.92	1.93	1.94
100	Reweighting	0.511	0.506	0.506	0.572	0.568	0.566
100	LSPIM	0.248	0.216	0.216	0.283	0.252	0.250
100	MI-LSPIM	3.75	3.52	3.51	3.74	3.53	3.55
1000	Classical ML	0.197	0.207	0.202	0.212	0.219	0.219
1000	MI	13.1	13.5	12.7	13.2	13.2	13.3
1000	Reweighting	14.7	14.7	14.7	16.5	16.4	16.4
1000	LSPIM	57.3	44.3	41.2	64.2	50.3	46.7
1000	MI-LSPIM	-	-	-	-	-	-

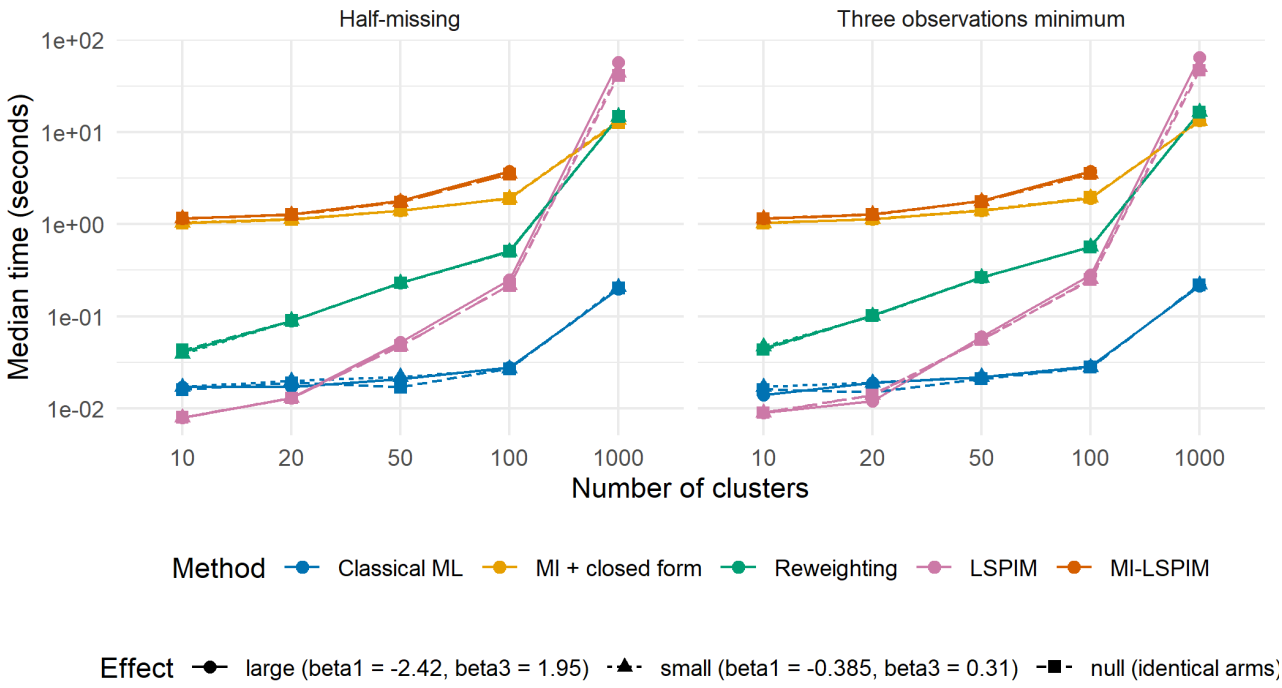


Figure 13: Median computation time per replicate for the log time trend (log scale).

Suggested improvements

1. Classical ML convergence: replace lme4's absolute gradient flag with a relative gradient check, to remove the false "not converged" at large n.
2. Currently, naive Wald t-test with N-2 df is used. This could be improved on.
3. MI-LSPIM is very conservative (type I error far below 0.05): investigate more imputations (m) or an alternative df or pooling.

Appendix: LSPIM and MI-LSPIM errors and warnings

Table 15: LSPIM and MI-LSPIM outcome per replicate (% of replicates, pooled over time trends and effect sizes). PSD = V repaired to the nearest PSD matrix; C1/C2 = leverage 1 in that clustering; zero col. = all-zero design column.

Method	n	Dropout	Converged	Warning (PSD)	Not conv.	Error (C1)	Error (C2)	Error (zero col.)
LSPIM	10	HM	0.0	79.0	0.3	10.1	9.7	0.8
LSPIM	10	3O	0.0	79.3	0.2	9.8	9.9	0.8
LSPIM	20	HM	0.0	99.9	0.0	0.1	0.1	0.0
LSPIM	20	3O	0.0	99.9	0.0	0.1	0.0	0.0
LSPIM	50	HM	98.0	2.0	0.0	0.0	0.0	0.0
LSPIM	50	3O	99.1	0.9	0.0	0.0	0.0	0.0
LSPIM	100	HM	100.0	0.0	0.0	0.0	0.0	0.0
LSPIM	100	3O	100.0	0.0	0.0	0.0	0.0	0.0
LSPIM	1000	HM	100.0	0.0	0.0	0.0	0.0	0.0
LSPIM	1000	3O	100.0	0.0	0.0	0.0	0.0	0.0
MI-LSPIM	10	HM	0.1	99.7	0.2	0.0	0.0	0.0
MI-LSPIM	10	3O	0.1	99.8	0.2	0.0	0.0	0.0
MI-LSPIM	20	HM	46.8	53.2	0.0	0.0	0.0	0.0
MI-LSPIM	20	3O	39.7	60.3	0.0	0.0	0.0	0.0
MI-LSPIM	50	HM	100.0	0.0	0.0	0.0	0.0	0.0
MI-LSPIM	50	3O	100.0	0.0	0.0	0.0	0.0	0.0
MI-LSPIM	100	HM	100.0	0.0	0.0	0.0	0.0	0.0
MI-LSPIM	100	3O	100.0	0.0	0.0	0.0	0.0	0.0

- LSPIM uses three clusterings of the pairwise pseudo-observations: by left subject (C1), by right subject (C2) and by pair (C3); the combined covariance is $V = V_{C1} + V_{C2} - V_{C3}$.
- Errors occur almost only at $n = 10$ (about 20% of replicates):
  - about 10% each: a cluster has leverage 1 in clustering C1 or C2, so the Ford-Westgate small-sample correction is undefined;
  - about 1%: no between-arm pairs at one or more late visits (one arm has no observations left there), so the corresponding design columns are all zero.
- Warnings: at $n = 10$ and 20 nearly every fit that does not error has to repair the combined covariance V to the nearest positive semi-definite matrix (negative eigenvalues or non-positive variances); about 1–2% of fits at $n = 50$, and none from $n = 100$.
- Not converged: rare (below 1% at $n = 10$, almost none from $n = 20$); the penalised GEE gives fitted probabilities numerically 0 or 1.
- MI-LSPIM has no errors in any replicate.
- MI-LSPIM warnings: the combined covariance V is repaired to a positive semi-definite matrix in 99.7% of fits at $n = 10$, 56.8% at $n = 20$, 0.01% at $n = 50$ and none at $n = 100$.
- MI-LSPIM not converged: 0.2% of fits at $n = 10$, one fit at $n = 20$ and none from $n = 50$ (0.05% over all fits).